

Unit 4: Mapping Earth's Surface

Reading: pages 51–65 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. On a map with a representative fraction of 1:24,000, one unit on the map represents
 - a) 24,000 units on the ground
 - b) 24,000 feet on the ground
 - c) 24,000 miles on the ground
 - d) one twenty-four-thousandth of a mile
2. What is the main advantage of a graphic scale (a scale bar) over other ways of showing scale?
 - a) It works with every map projection
 - b) It shows elevation as well as distance
 - c) It is more mathematically precise
 - d) It remains true when the map is shrunk or magnified
3. On a topographic map, contour lines drawn very close together indicate
 - a) a steep slope
 - b) the edge of a body of water
 - c) an error in the survey
 - d) relatively flat terrain
4. The Mercator projection is *conformal*, which means that it
 - a) preserves area but distorts shape
 - b) preserves shape and angle but strongly distorts area
 - c) preserves distance from a single central point
 - d) preserves shape, area, and distance equally
5. Which developable surface is most often used to map the Polar Regions?
 - a) a cylinder
 - b) a cone
 - c) a plane
 - d) an ellipsoid

6. In the Public Land Survey system, a township is an area measuring
- a) 1 mile on a side
 - b) 6 miles on a side
 - c) 36 miles on a side
 - d) 40 acres

Part B · True or False

2 questions × 1 point = 2 points

7. The *extent* of a map describes the smallest unit that is mapped.
TRUE / FALSE
8. No map projection can preserve all properties of the globe at once.
TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. Name the three problems a cartographer runs into when flattening a globe. Which one do developable surfaces eliminate entirely, and why is that the one worth eliminating?
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10. Explain the difference between a *conformal* projection and an *equal-area* projection, and give one situation in which you would choose each.
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